PES UNIVERSITY

CENTRE FOR INNOVATION & ENTREPRENEURSHIP

Repo6uard

AUTOMATIC CODE VULNERABILITY DETECTION FOR OPEN SOURCE PROJECTS

“because insecure code costs more”

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Problem Statement

What is the Problem your startup team is trying to solve?

1. Who has the problem?

- Maintainers
- Developers
- Companies
- Security teams
- End users

2. What does it cost them?

- Legal penalties & compliance fines
- Emergency patching & downtime
- Indirect Costs
- Loss of public trust
- Delayed releases

3. How often does it happen?

- Code commit
- Dependency update
- Open-source library upgrade
- New CVEs are published daily

This is not a one-time issue — it's a persistent risk.

4. Specific pain points:

- Manual code review is time consuming and not reliable
- Vulnerabilities are detected after deployment usually
- Lack of consistent security standards across open-source repositories

Existing Solutions and Competition

- **Existing Approaches/Solutions & Competition:**
  - Rule-based security scanners (static analysis)
  - Dependency vulnerability scanners
  - Manual security audits
  - Basic CI/CD security checks
- **Drawbacks of Current Approach (Mostly Reactive and Rule-Driven):**
  - High number of false positives
  - No understanding of code context or intent
  - Limited detection of unknown or logic-based vulnerabilities
  - Minimal or no automated fix suggestions

Your Solution (Product/Service/Offering)

What is the proposed solution you plan to offer? What is novel?

- **Superior Need Fulfillment:** Reduces false positives and catches unknown threats through context-aware AI.
- **Value Proposition:** Prevents vulnerabilities before deployment to slash audit costs and build software trust.
- **Specific Results:** Delivers faster detection and improved compliance readiness with fewer irrelevant alerts.
- **Feasibility:** Built on proven AI and static analysis tech with seamless CI/CD pipeline integration.
- **Innovation:** Features a unique hybrid AI detection engine paired with automated remediation suggestions.

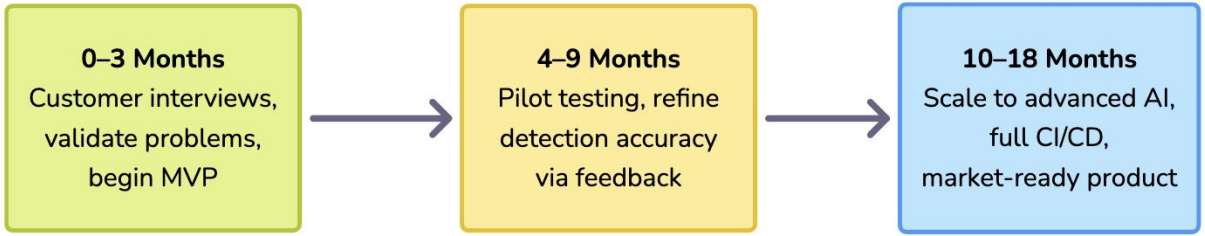
Market Opportunity

Market Opportunity & Timing

- **Target Customers:** Securing government departments, open-source-heavy enterprises, and overextended DevSecOps teams.
- **Desirability:** Prevents high-cost breaches while automating routine audits to solve the critical talent shortage.
- **Market Opportunity (TAM):** The global application security market is projected to reach **$43.87 billion** this year.
- **Market Opportunity (SAM):** The DevSecOps segment is valued at **$11.49 billion** as security "shifts left" in 2026.
- **Market Opportunity (SOM):** India’s cybersecurity product revenue is on track to reach **$6 billion** in 2026.
- **Why Now? (Trends):** AI-driven threats now move at machine speed, requiring AI-native autonomous defenses to keep pace.

Proposed Implementation Plan

High-level plan for taking this forward



- **Semester Success:** A functional MVP that successfully scans real repositories and receives validation from mentors.
- **Resources Needed:** High-compute GPU/cloud access, open-source vulnerability datasets, and industry mentor feedback.
- **Team Execution:** Agile, milestone-based development with clear roles leveraging skills in Python, C, and security auditing.

RESEARCH AND REFERENCES

1.https://finitestate.io/blog/how-to-identify-vulnerabilities-in-open-source-code